

# SREE GNANESH DOMMETI

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## Education

### University of Maryland, College Park

*M.S. in Applied Machine Learning; GPA: 3.9 / 4.0*

May 2026

College Park, MD

### National Institute of Technology, Calicut

*B.Tech. in Computer Science and Engineering; GPA: 8.21 / 10*

May 2022

Calicut, India

## Experience

### Qualcomm

July 2022 – July 2024

*Software Engineer*

Hyderabad, India

- Owned production LTE Layer 3 RRC Cell Selection and Security modules; designed fuzzing strategies for malformed OTA messages and fake base-station behavior.
- Delivered feature enhancements for CRS/AUX capture, MPS priority indication, OMPD refactoring, and multivector/EFS logging, improving diagnostics and carrier/OEM compliance.
- Built Python automation tools for changelist assignment, regression tracking, priority assignment, and RRC ownership routing; reduced manual engineering effort by 40%.

### FROOT Lab, University of Maryland

November 2025 – Present

*Research Assistant, Advisor: Alan Zaoxing Liu*

College Park, MD

- Worked on ProjectASAP, implementing research-backed systems components for workload analysis, sketch planning, streaming summaries, and approximation-aware query serving.
- Built software paths across Prometheus, Arroyo, Kafka, and Rust query services, helping move the project from research prototype toward reproducible telemetry benchmarks.
- Built NetArena MAL Agent tooling for capacity-planning tasks with Dockerized LiteLLM prompts, output validation, retries, and local submissions.

### GAMMA Lab, University of Maryland

January 2026 – May 2026

*Research Assistant, Advisor: Dinesh Manocha*

College Park, MD

- Contributed to ongoing grounded MLLM/video reasoning evaluation tooling, checking intermediate reasoning steps against grounded visual/textual evidence.

### Connyct

June 2025 – August 2025

*ML Engineer Intern, CampusAI*

College Park, MD

- Built recommendation services and Elasticsearch retrieval for a campus-event discovery backend; normalized event feeds and served personalized results at sub-300 ms p95 retrieval and sub-600 ms p95 end-to-end latency.
- Containerized services with Docker, deployed on AWS EC2, integrated MLflow tracking, and added offline ranking evaluation with Recall@K, NDCG@K, MRR@K, cold-start recall, and long-tail coverage.

## Projects

### eXpOS - Experimental Operating System Kernel | C, XSM Assembly, Paging, System Calls

- Implemented process, memory, file, device, and resource-management modules on the XSM simulator, enabling multitasking, timer-driven context switching, and UNIX-like process execution.
- Built demand paging with page tables and page-fault handling; implemented 19 interrupts/system calls including fork, exec, wait, signal, semaphores, terminal I/O, and disk operations.

### CallIntel Agent - Audio Support-Call QA | Python, Cloud Run, Pub/Sub, BigQuery, Bedrock Payloads, OpenTelemetry

- Built an audio-to-RAG support QA prototype that ingests call-audio manifests, redacts transcript PII, retrieves transcript/policy evidence, and returns structured answers with citations and tool traces.
- Added GCP artifacts, Gemini/OpenAI/Claude/Bedrock provider payload validation, and Datadog-compatible OpenTelemetry spans/metrics for retrieval, provider routing, citation quality, and latency.

### ProjectASAP - Streaming Telemetry Query System | Python, Rust, Arroyo, Kafka, Prometheus, Docker

- Implemented workload-analysis and sketch-planning services for high-cardinality telemetry queries, using configurable sketch choices and transparent fallback to Prometheus for unsupported query paths.
- Connected Prometheus ingest, Arroyo streaming jobs, Kafka sketch transport, and Rust query-serving components into an approximation-aware telemetry pipeline.

## Skills

**Languages:** C, C++, Python, SQL, JavaScript, Rust, Dart

**Backend / Systems:** FastAPI, Node.js, Express.js, REST APIs, structured JSON outputs, Linux, paging/system calls, concurrency, debugging, protocol stacks, fuzz testing

**Data / Infra:** PostgreSQL, MySQL, MongoDB, Elasticsearch, Redis, Kafka, Prometheus, Grafana, Docker, Kubernetes, AWS EC2/ECS/S3/Lambda, Bedrock payloads, GCP Cloud Run/Pub/Sub/BigQuery/IAM, Terraform

**Tools:** Git, GitHub Actions, Jenkins, Perforce, Jira, Postman, PyTorch, MLflow, DVC, OpenTelemetry, Datadog-compatible OTLP specs